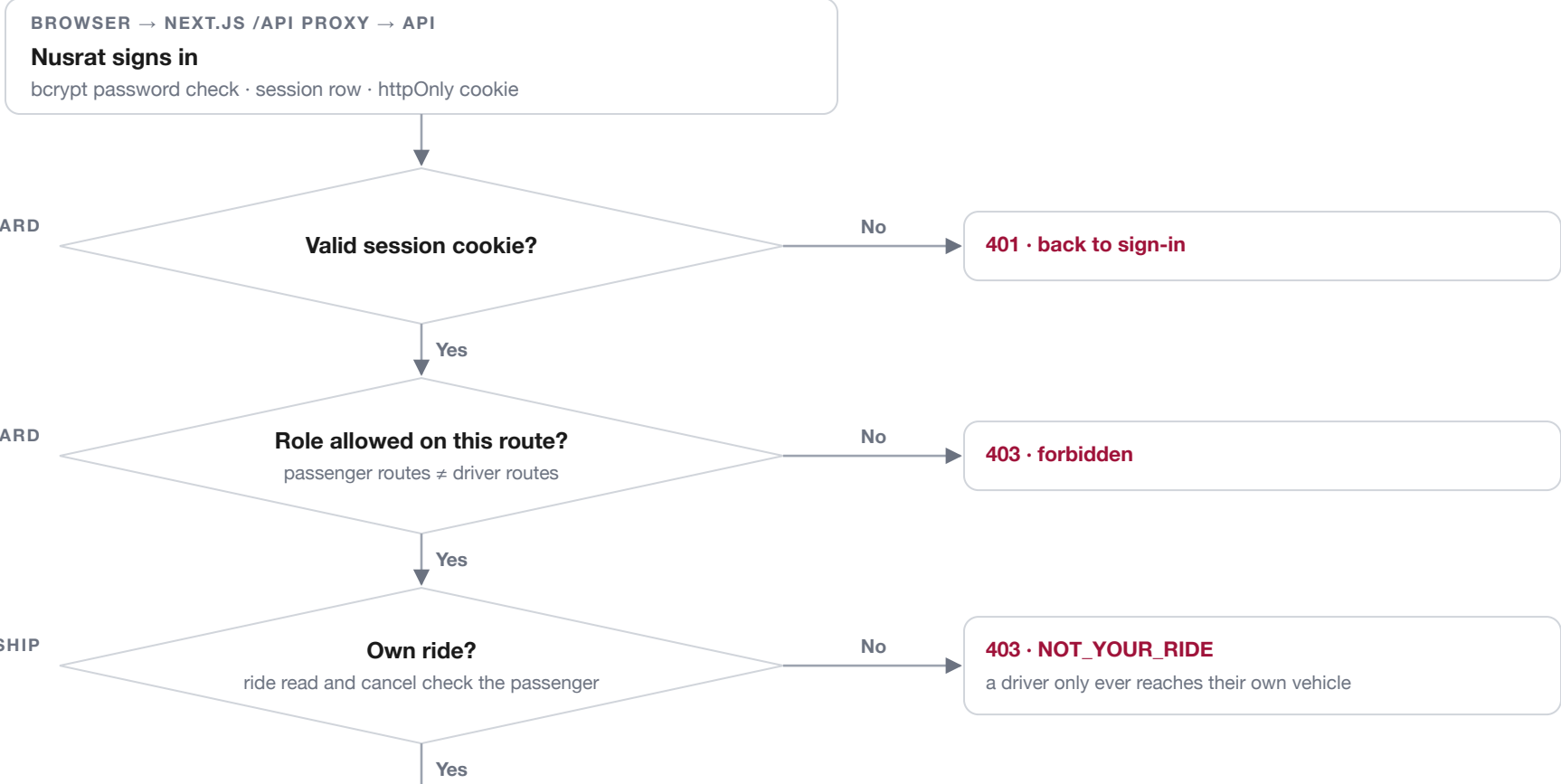


Dhaka Tesla Pool · Ride Flow

How one ride moves through the system · a Tesla picks people up along its route · side notes cover concurrent and edge cases

1 SIGN IN



Sign up

Passengers only; drivers are seeded
Unique email (stored lower case)
Password hashed with bcrypt
Sign-up and login: 5 per minute

2 REQUEST A RIDE



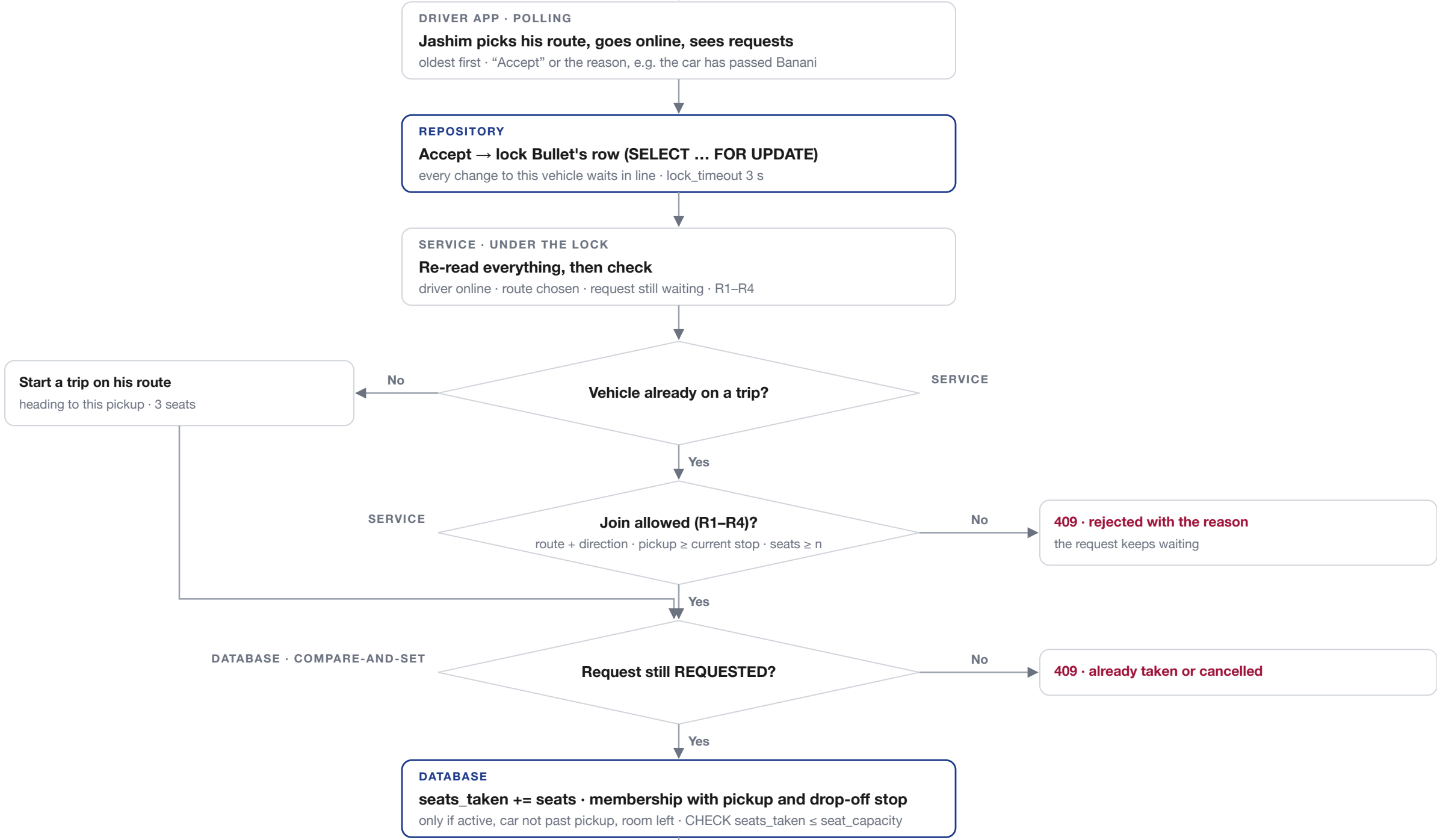
Routes (R1)

6 routes: 3 lines, both directions
Uttara → Banani → Mohakhali →
Gulshan 1 → Gulshan 2 → Bashundhara
Banani → Uttara: other direction X

Fare model

(Tk 30 + direct km × Tk 15) × seats
–20% if you shared at least one hop
Estimate = solo fare, never exceeded
Locked at drop-off

3 MATCHING



Last seat, many riders at once

Nusrat holds 2 of Bullet's 3 seats
8 riders ask for the last one
All wait for Bullet's lock
Exactly 1 wins · 7 keep waiting

No trip yet, two accepts at once

First accept starts the trip
Second waits, then joins it
One trip, two passengers

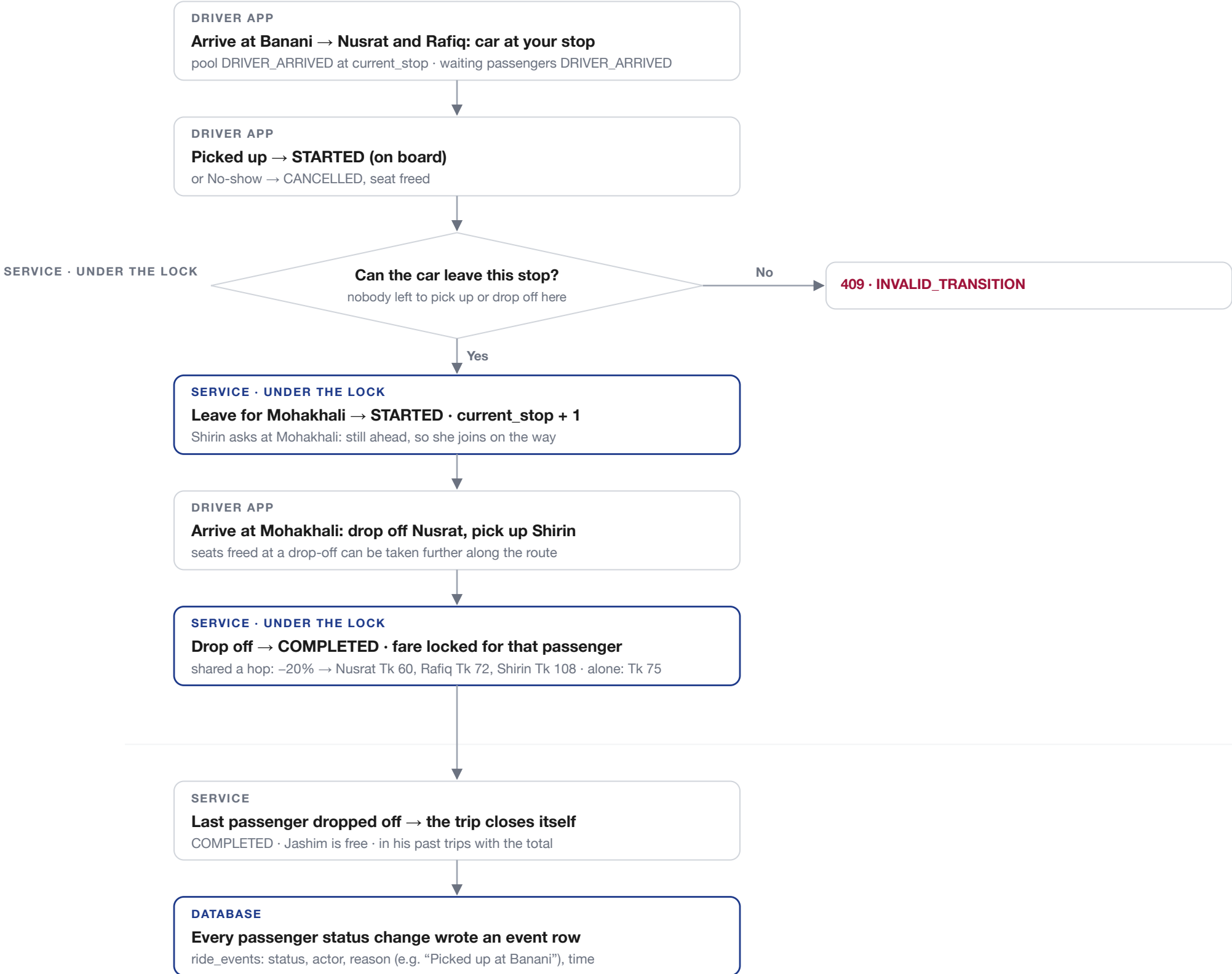
Retries and timeouts

Lock wait > 3 s → 503 BUSY
Auto-join is best effort: if busy,
the ride just keeps waiting

Database guarantees

seats_taken ≤ seat_capacity
pickup_stop < droptoff_stop
1 active pool per vehicle
1 active membership per ride
1 active ride per passenger

4 THE TRIP, STOP BY STOP



Online / offline

Needs a route; changed under the lock
Route changes only between trips
Offline refused while a pool is active
Accept refused while offline

Leave vs join at the same stop

Car at Mohakhali, Shirin asks there
Both wait for Bullet's lock
Join first → the car waits for her
Leave first → stop is behind: she waits

What the passenger sees

Polls her ride every 3 s (history 10 s)
The route with the car on it
Own status and own fare only
Co-riders' first names only

Cancellation rules

Passenger: until picked up
→ seats freed at once
Driver at the stop: no-show
Driver: before the first pickup →
passengers back to REQUESTED
Last one out → the trip closes

5 END OF TRIP